Chapter 21 - Statics

#Mechanics

Mechanics is the branch of applied mathematics that studies the motion of bodies and the forces acting on them. It can broadly be divided into two parts: Statics and Dynamics.

Statics	Dynamics
Statics is the branch of mechanics that deals with the study of bodies at rest or at equilibrium.	Dynamics is the branch of mechanics that deals with the study of bodies at motion.
It considers all forces and moments are balanced, maintaining stability.	It considers forces are unbalanced, producing motion or change in motion.

Fundamental Quantities of Mechanics

The fundamental quantities of mechanics are mass (M), length (L) and time (T). All the quantities used in mechanics are defined on the basis of these quantities.

Basic Terms

Body	A body is a portion of matter having a finite shape and size and occupying certain space.
Rigid Body	A rigid body is a body in which the distance between any two points remains the same and does not alter with time. In other words, a body that does not deform is termed as a rigid body.
Particle	A particle is a portion of matter which is so small in size that the distance between any two of its parts can be neglected. It does not change its rigidity under the action of external forces.
Mass	Mass is the amount of matter contained in a body. The SI unit of mass is kilogram (kg).

#Force

A force is an external agency (push or pull) which changes or tries to change the state, direction, speed, shape, or equilibrium of a body.

Representation of a Force

A force is represented as a vector quantity $\vec{F}$ possessing both magnitude and direction, commonly depicted by a straight arrow where length indicates strength and the arrowhead indicates direction.

Types of Force

Forces are classified into following types:

Attraction	The force exerted by one body on another body with no use of visible medium, is known as the attraction.
Reaction	The force exerted by a body in response to an action force applied by another body, is known as the reaction.
Tension	The force transmitted axially through a string when it is pulled, is known as the tension.

Units of Force

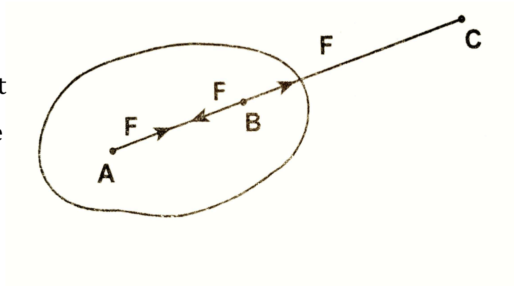
×	SI System	CGS System	FPS System
Relative Unit	kg. wt.	gm. wt.	1 lb. wt.
Absolute Unit	Newton (N)	dyne	poundal (pdl)

Equilibrium of Forces

Equilibrium of forces is the state when the net external force acting on an object is zero, resulting in balanced forces where the object remains at rest (static equilibrium) or moves at a constant velocity (dynamic equilibrium).

Principle of Transmissibility of Forces

If a force acts at any point of the rigid body, it may be considered to act at any other point in its line of action provided that this latter point be rigidly connected with the body.

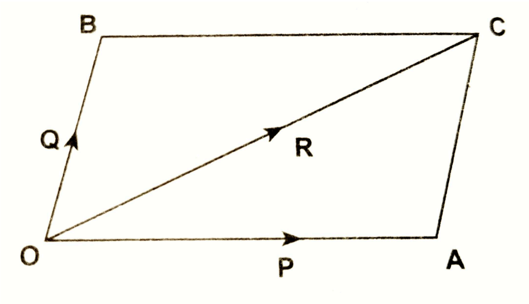


#Composition of Force

If two or more forces act upon a body and a single force is obtained such that the effect of the single force upon the body is the same as the effects due to two or more forces, then the single force is called the resultant of the given two or more forces and each force of the two or more forces is called the component of the resultant force.

#Parallelogram of Forces

If the two forces acting at a point can be represented in magnitude and direction by the adjacent sides of a parallelogram drawn through the point then their resultant is represented in magnitude and direction by the diagonal of the parallelogram drawn through the same point.



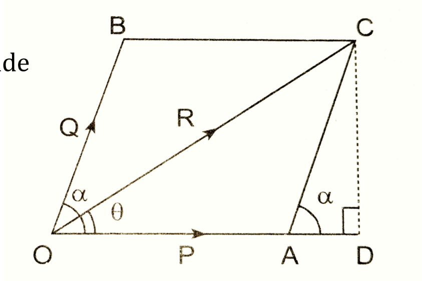
#Resultant of two given forces

If two forces P and Q are acting at a point O at an angle α then the magnitude of their resultant is given by

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\alpha}$$

If the angle made by the resultant with P is θ then its direction is given by

$$\tan\theta = \frac{Q\sin\alpha}{P + Q\cos\alpha}$$

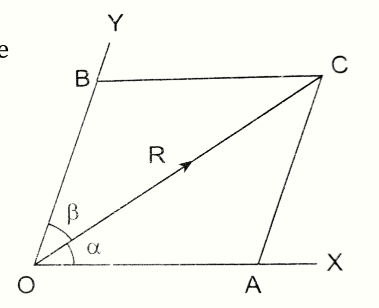


Condition	Resultant	
	Magnitude	**Direction**
$\alpha = 90°$	$R = \sqrt{P^2 + Q^2}$	$\tan\theta = \frac{Q}{P}$
$P = Q$	$R = 2P\cos\frac{\alpha}{2}$	$\theta = \frac{\alpha}{2}$
$\alpha = 0°$	$R = P + Q$	$\theta = 0°$
$\alpha = 180°$	$R = P - Q$	$\theta = 180°$
$P = Q = R$	$R = P$	$\theta = 60°$

#Resolution of Forces

A force may be resolved into two components in an infinite number of ways in any two assigned directions. The two cases of resolution of a given force are as follows:

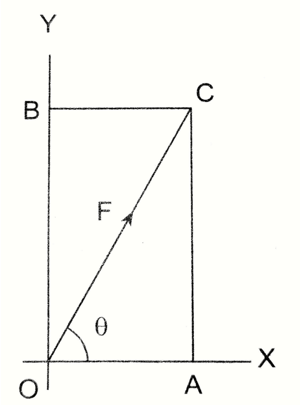
Resolution of a given force into two components along two given direction



If a force R is resolved in two components P and Q such that the given force makes angles α and β with the respective components then the magnitude of the component is given by

$$P = \frac{R \sin \beta}{\sin (\alpha + \beta)} \quad \text{and} \quad Q = \frac{R \sin \alpha}{\sin (\alpha + \beta)}$$

Resolution of a given force into two components at right angles to one another

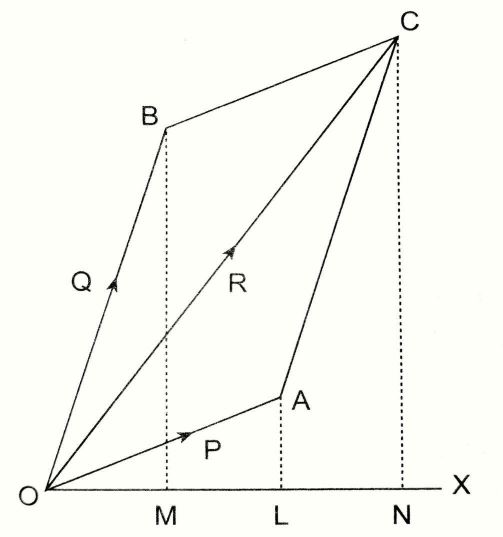


If a force F is resolved into two perpendicular components perpendicular such that it makes angle θ with the horizontal component then

$$F_x = F \cos \theta \qquad \text{(Horizontal component)}$$
$$F_y = F \sin \theta \qquad \text{(Vertical component)}$$

Theorem

"The algebraic sum of the resolved parts of any given forces acting at a point and in the given directions, is equal to the resolved part of their resultant in the same direction."



If the force P, the force Q and their resultant R makes angles α, β and θ with the horizontal axis respectively then

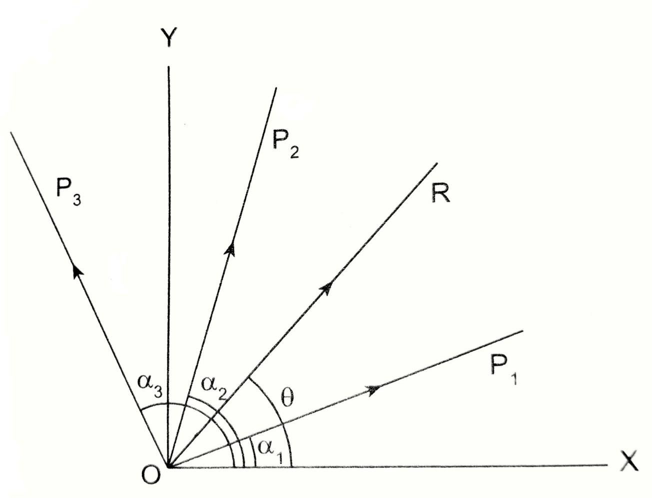
$R \cos \theta = P \cos \alpha + Q \cos \beta$ and $R \sin \theta = P \sin \alpha + Q \sin \beta$

Corollary

If $P, Q, R, S, \ldots$ are forces acting at O inclined at angles $\alpha, \beta, \gamma, \delta, \ldots$ and F is their resultant inclined at an angle θ with the positive x-axis then

$$F_x = F \cos \theta = P \cos \alpha + Q \cos \beta + R \cos \gamma + \ldots$$
$$F_y = F \sin \theta = P \sin \alpha + Q \sin \beta + R \sin \gamma + \ldots$$

Resolution of coplanar forces acting at a point



If the coplanar forces $P_1, P_2, P_3, \ldots$ are acting at a point O and inclined at angles $\alpha_1, \alpha_2, \alpha_3, \ldots$ with positive x-axis respectively and their resultant is R which makes an angle θ with the positive x-axis then

$$R_x = R\cos\theta = P_1\cos\alpha_1 + P_2\cos\alpha_2 + P_3\cos\alpha_3 + \ldots$$

$$R_y = R\sin\theta = P_1\sin\alpha_1 + P_2\sin\alpha_2 + P_3\sin\alpha_3 + \ldots$$

Magnitude of Resultant

$$R = \sqrt{R_x^2 + R_y^2}$$

Direction of Resultant

$$\tan\theta = \frac{R_y}{R_x}$$

Forces in Equilibrium

Forces acting on a particle or point are said to be in equilibrium if their resultant is zero.

If $F_1, F_2, F_3, \ldots, F_n$ are n coplanar forces acting on a particle which is in equilibrium then

$$\Sigma F_i = 0$$

$$\Rightarrow \Sigma X_i = \Sigma Y_i = 0$$

In case of non-coplanar concurrent forces, the condition of equilibrium is

$$\Sigma X_i = \Sigma Y_i = \Sigma Z_i = 0$$